Leire Lasa

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Education

University of Toronto Sep 2024 – Apr 2029
B.A.Sc. in Mechanical Engineering + PEY Co-op *Toronto, ON*

- **Cumulative GPA: 3.76** (Dean's Honors Scholar, U of T Engineering International Scholar Award)

Experience

Medical Imaging Service Intern May 2026 – Jul 2026
Electronica Medica, S.A. *Panama City, Panama*

- Performed 30+ maintenance procedures on imaging systems following PM checklists, with visual and self-diagnostic software inspections and service reports
- Resolved mechanical, electronic, and image-quality issues on mammography systems (filter and optical fiber replacement, dead-pixel correction, brake repairs); repaired ultrasound, X-ray, and fetal monitors
- Reverse engineered a damaged hospital-bed button from caliper measurements; modeled in SolidWorks for 3D printing

Engineering Operations Intern Jun 2025 – Aug 2025
Mantra Global Corporation, S.A. *Panama City, Panama*

- Inspected BOMs and engineering drawings for misplaced holes, incorrect units, and part labelling, and coordinated corrective actions through root-cause analysis to prevent beam-building errors and production delays
- Developed a Google Sheets beam tracking system for simultaneous 33-axis-3-floor and 26-axis-1-floor projects, addressing duplicate beam production and overcrowded storage
- Supervised welders, builders, sandblasters, and painters to deliver beams for two simultaneous projects; evaluated electrode and wire welding fillets to ensure quality compliance and reduce faulty beam deliveries

Quality Assurance and Industrial Safety Supervisor Feb 2024 – May 2024
Gaspro Panama, S.A. *Panama City, Panama*

- Executed workplace safety audits and delivered 10+ training programs, coordinating with industrial hygiene, fire safety, and environmental agencies to maintain ISO 9001:2015 compliance
- Improved gas cylinder storage and inventory flow with FIFO practices, aligning production volumes with customer orders

Projects

UTWIND: Small-Scale Wind Turbine | *SolidWorks, Circuit Design, Current sensors* Sep 2026 – Present

- Designed an adjustable swashplate pitch-control mechanism ($\pm 15^{\circ}$) improving low-wind startup and aerodynamic braking
- Built current-sensor test circuitry with shunt resistors and decoupling capacitors for signal stability and accuracy

UTBIOME: Emergency Triage System | *Team Lead, SolidWorks, PrusaSlicer, I^2C* Oct 2025 – Present

- Reverse engineered a commercial automatic wrist blood-pressure monitor to understand its pump and valve inflation mechanism, and researched the Korotkoff sound method to determine how a MAX30102 heart-rate sensor could communicate with the pressure sensor to automate cuff inflation
- Designed a manual backup cuff using the same principles, with user prompts indicating when to inflate or allow pressure to release, in case the automated system fails
- Benchmarked existing oximeter clamp designs for compatibility with the MAX30102, then prepared and 3D printed the selected clamp using PrusaSlicer for slicing

UTBIOME: ACL Rehabilitation Knee Sleeve | *SolidWorks, sEMG* Oct 2025 – May 2026

- Developed a rehabilitative ACL knee sleeve in SolidWorks using 3D sketching and advanced surface modeling (sweeps, offset, knit, thicken) to replicate leg geometry and support ANSYS interface-pressure simulations
- Evaluated 10+ conductive textiles for sEMG integration via peer-reviewed literature, improving sensing and breathability
- Interpreted ISO 10993-1:2025 to select hypoallergenic materials for a Class A skin-contact device

Cinematography Robot Arm | *SolidWorks, Kinematic Analysis* Sep 2025 – Dec 2025

- Led a team of four in designing a 7-DOF robotic arm for intermediate hobbyist cinematographers, focusing on repeatability and cost-effectiveness; benchmarked 5+ commercial systems across 20+ metrics
- Performed torque and force analyses on arm and wrist mechanisms to validate motor selection under maximum load
- Designed base, shoulder, arm, and wrist/end-effector subassemblies in SolidWorks using splines, sweeps, lofts, revolves, Hole Wizard, and feature tools (fillets, chamfers, shells, patterns, splits)
- Assembled subassemblies using standard, advanced, and mechanical mates, integrating Toolbox components and supplier CAD models to validate fit, motion, and subsystem interfaces

Technical Skills

Design and Simulation: SolidWorks (CSWA certified), SolidWorks CAM, MATLAB, LTSpice, LabVIEW
Programming and Data: Python (Spyder, Jupyter), Minitab, Excel, Google Sheets
Manufacturing: Lathe, CNC milling, drill press, grinding (George Brown Machining Course), 3D printing (PrusaSlicer)
Languages: Spanish (Native), English (Fluent)